Does being happy make you more likely to recycle more? (in London boroughs from 2011-2023)

Abstract

While a new recycling program has just rolled out across England on March 31st, unifying all waste collection across the whole country and adding a 4th bin for paper and card, we want to look at not just policy factors into why people recycle. We want to look at happiness as a driver for recycling habits in London boroughs from 2011 – 2023. Using regression models, we aim to predict that higher happiness levels correlate to higher recycling rates while controlling for other factors. Results reveal that a simple corelation does exist however once other factors are added the validity of said corelation gets blurred. It also reveals that housing density and financial stability are the main predictors and budget and policy are weaker. Meaning to address recycling rates we may need to think about infrastructure and financial security rather than relying on the budget of recycling programs.

Intro

Does happiness predict household recycling rates across the 32 London boroughs.

Hypothesis: Happiness is a positive predictor of household recycling rates.

Null hypothesis: Happiness is not a predictor of household recycling rates.

Primarily, we centred on this question due to our current year's focus on environmental change and inequality. We found solid datasets on happiness and household recycling rates across government and local authority waste management records. We began to think about this question in the form of littering, does being in a good mood make you want to protect the environment. We thought this could be an interesting study.

Data

Happiness

- Recycling Rates (RR)
    - Housing Density
    - Net recycling Expenditure per borough per capita
    - Earnings residence
    - Waste collection cost

<u>Happiness</u>

personal-well-being-borough.xlsx

This data comes from the Office for National Statistics which is a reputable source.

This is our independent variable or predictor. The data we are measuring everything against.

Recycling Rates

Household-rcycling-borough.csv

This data comes from the London data store, which is taken from the UK Department for Environment, Food and Rural Affairs.

This is the dependent variable. We are using happiness as a predictor for recycling.

Housing density

Housing-density-borough.csv

- Land Area & Population - Borough (CSV Version)

This data comes from the London data store through the office for national statistics.

This variable controls for the structural challenges of urban environments. Households with less space don't have as much room for recycling infrastructure.

Net recycling Expenditure per borough per capita

Revenue outturn 2011-2023 [datasets] - in teams shared

This data is from waste data flow from gov.uk.

This variable controls for the investment that each borough has for recylcing. Being that more recycling infrastructure has more recycling.

Earning residence

Earnings-residence-boroughs.xlsx

- Total, weekly-Table 1

This data comes from official census and labour market statistics which is a subsect of the office for national statistics.

This variable controls for socioeconomic reasons for recycling. Being that people with higher financial security have more time to prioritize recycling.

Waste collection cost

Revenue outturn 2011-2023 [datasets] - in teams shared

This data is from waste data flow from gov.uk.

This variable controls for the general operational cost of waste management per borough. Distinguishing between specific cost of recycling management and removing waste.

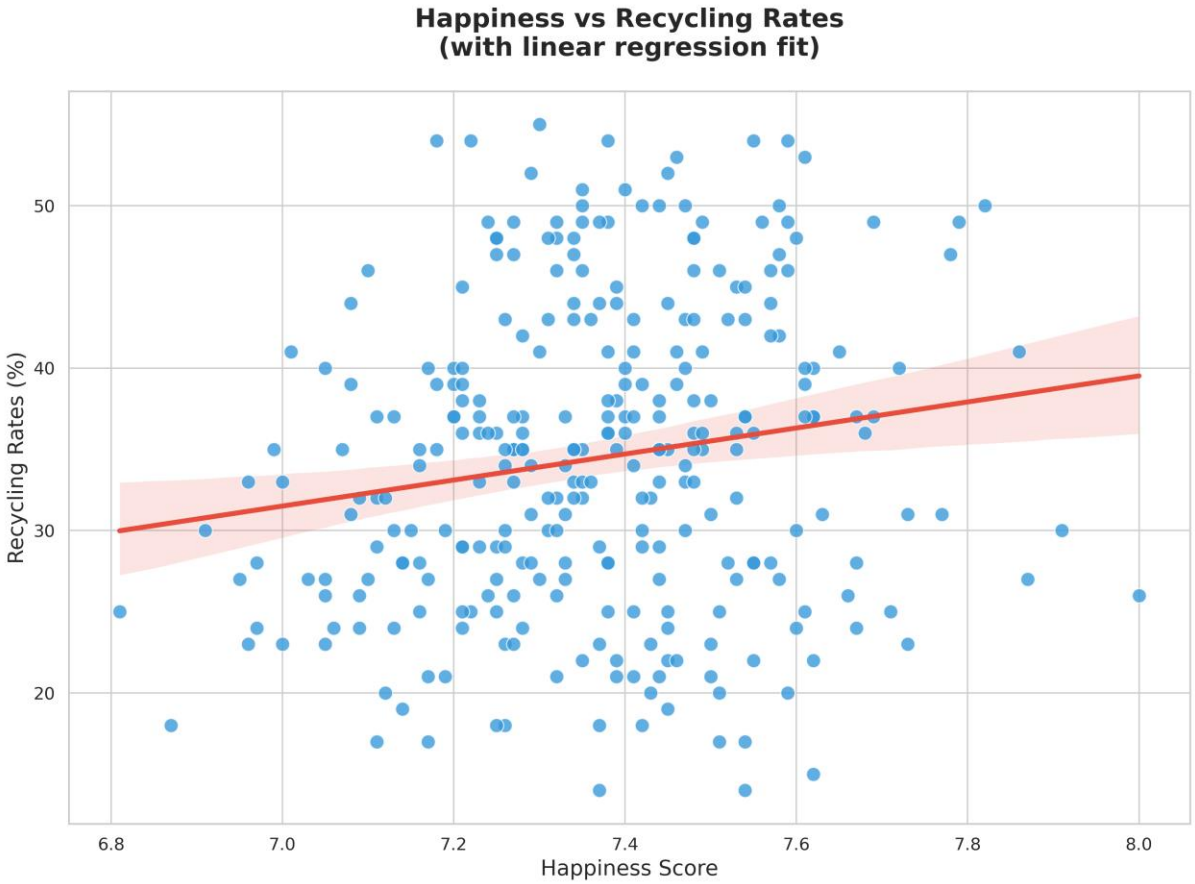
Model

1. [Min] Comparing happiness and Recycling rates.

2. [Max] Comparing happiness and Recycling rates with housing density, Net recycling Expenditure, Earnings residence, and Waste collection cost to control for different recycling rates.

3. [Robustness Check] Comparing happiness and Recycling rates with housing density, Net recycling Expenditure, Earnings residence, and Waste collection cost to control for different recycling rates. BUT using happiness inequality rather than happiness mean scores.

Results

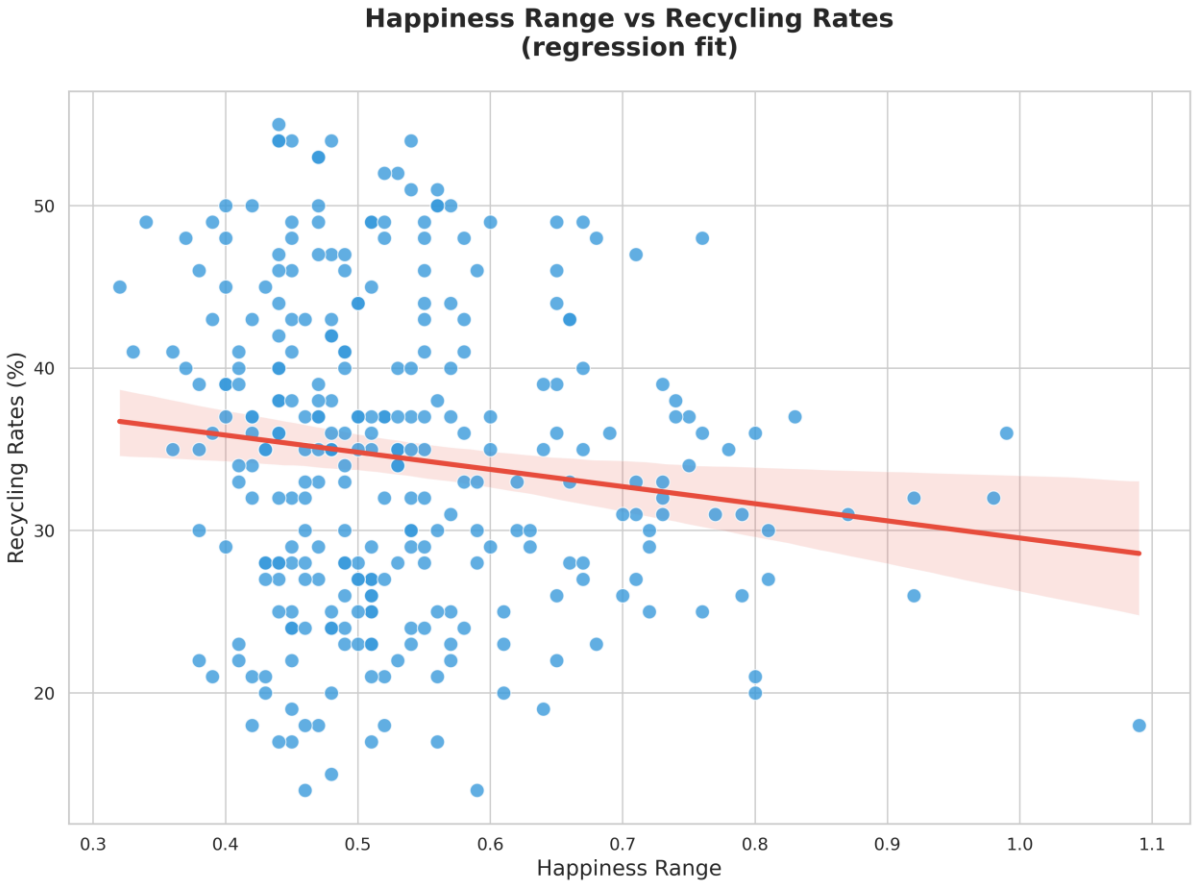


- Comparing happiness and Recycling rates.

The initial model suggests that there is significant positive relationship between happiness and recycling rates with a p value of 0.004. Therefor we can reject the null hypothesis that happiness and no effect on recycling. HOWEVER, looking at the r squared is 0.027. This means the less than 3% is explained by the variation in recycling. Which means that happiness is probably not a primary driver of recycling.

- Comparing happiness and Recycling rates with housing density, Net recycling Expenditure, Earnings residence, and Waste collection cost to control for different recycling rates.

When controlling for factors the significance drops significantly and it becomes statistically insignificant with a p value of 0.391. population density and income are big predictors for this model, meaning that higher density correlates to lower recycling rates and higher income earning correlates to higher recycling rates. The jump in r squared (0.411) compared to the first model tell us that this model is more effective at explaining the recycling rates.



- Comparing happiness and Recycling rates with housing density, Net recycling Expenditure, Earnings residence, and Waste collection cost to control for different recycling rates. BUT using happiness inequality rather than happiness mean scores.

This model replaces the happiness with the happiness range of each borough. More specifically the gap between the upper and lower limits of the happiness level. This gets us more variation of the happiness predictor because the previous data was a mean per borough and they were all similar, even though some areas in the borough could be lower than others. The results show that happiness inequality isn't a significant predictor in recycling rates in London boroughs ($p = 0.173$). this solidifies the previous model that the main dominate predictors are population density and residence earning, not happiness.

Discussion

While the initial model was statistically significant the second model could lead us to understand that there wasn't a casual effect of happiness and recycling. Perhaps that happiness can be predicted by the main dominant factors of population density and income. Or considering the p value was so small, something else was going on with the data that interacted with the results.

While we did not manage to prove our hypothesis, we learned some things. Mainly that population density and income are major predictors of our recycling rates in London. Also, what is interesting is that recycling expenditure and waste cost were not a strong indicator of recycling rates. Meaning that throwing money into a recycling program might not have an impact on the environment, that we must look closer and what really makes people recycle.

So maybe the country's new recycling program won’t be so impactful as we would hope.